

cs1231s lecture notes

Types of statement

- **Universal:** true for all elements in set
- **Conditional:** if P_1 is true, another P_2 is also true
 - **Universal conditional:** for all a , if a match P_1 , then a match P_2 .
- **Existential:** at least 1 thing in a set is true
 - **Universal existential:** first part is universal, second part is existential
 - **Existential universal:** first part is existential, second is universal
- **Necessary:** It is not enough to prove $p \rightarrow q$ although p being true is one of many conditions needed to prove q .
- **Sufficient:** It is enough to prove $p \rightarrow q$ with p being true means q must be true.

Operations

- **Closure:** An operation on elements of a set produces a result that is also in the set.
- **Commutativity:** The order of the operands doesn't affect the result.
- **Associativity:** The way operands are grouped doesn't affect the result.
- **Distributivity:** One operation distributes over another.
- **Trichotomy:** For any two elements a and b in a totally ordered set, exactly one holds: $a < b$, $a = b$, or $a > b$.

Notation

priority	symbol	name
1	$\sim$	not (negation)
2	$\wedge$	and (conjunction)
2	$\vee$	or (disjunction)
3	$\rightarrow$	then/implies

$$a|b \rightarrow b = ac$$

If/then properties

Operations on statement

- **Negative:** the negative of $p \rightarrow q$ is $p \wedge \sim q$
- **Converse:** $p \rightarrow q$ is the converse of $q \rightarrow p$
- **Inverse:** $p \rightarrow q$ is the inverse of $\sim p \rightarrow \sim q$
- **Contrapositive:** $p \rightarrow q$ is the contrapositive of $\sim q \rightarrow \sim p$

Decomposition Rules

- **Biconditional Law:** $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
- **Implication:** $p \rightarrow q \equiv \sim p \vee q$
- **Contrapositive Law:** $p \rightarrow q \equiv \sim q \rightarrow \sim p$

iff

$$a \text{ iff } b \equiv a \rightarrow b \wedge b \rightarrow a$$

Soundness

An argument is **sound** if it is both **true** and **valid**.

Truth

1. Check if all premises are true.
 - If all true, argument is true and vice-versa.

Validity

Assume premises are true. Is the conclusion true?

1. Construct a truth table showing the truth values of all the premises and the conclusion.
2. A row of the truth table in which all the premises are true is called a **critical row**.
 - If there is a critical row in which the conclusion is false $\rightarrow$ the argument form is invalid.
 - If the conclusion in every critical row is true $\rightarrow$ the argument form is valid.

Rules of inference

Rule of Inference	Logical Form
Modus Ponens	$p \rightarrow q, p \vdash q$
Modus Tollens	$p \rightarrow q, \sim q \vdash \sim p$
Generalization	$p \vdash p \vee q; q \vdash p \vee q$
Specialization	$p \wedge q \vdash p; p \wedge q \vdash q$
Conjunction	$p, q \vdash p \wedge q$
Elimination	$p \vee q, \sim q \vdash p; p \vee q, \sim p \vdash q$
Transitivity	$p \rightarrow q, q \rightarrow r \vdash p \rightarrow r$
Proof by Division Into Cases	$p \vee q, p \rightarrow r, q \rightarrow r \vdash r$
Contradiction Rule	$\sim p \rightarrow \text{false} \vdash p$

Logical equivalences

Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative laws	$p \wedge q \wedge r \equiv (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$p \vee q \vee r \equiv (p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity laws	$p \wedge \text{true} \equiv p$	$p \vee \text{false} \equiv p$
Negation laws	$p \vee \sim p \equiv \text{true}$	$p \wedge \sim p \equiv \text{false}$
Double negative law	$\sim \sim p \equiv p$	
Idempotent laws	$p \wedge p \equiv p$	$p \vee p \equiv p$
Universal bound laws	$p \vee \text{true} \equiv \text{true}$	$p \wedge \text{false} \equiv \text{false}$
De Morgan's laws	$\sim (p \wedge q) \equiv \sim p \vee \sim q$	$\sim (p \vee q) \equiv \sim p \wedge \sim q$
Absorption laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negation of true and false	$\sim \text{true} \equiv \text{false}$	$\sim \text{false} \equiv \text{true}$

- Universal Instantiation: If some property is true of everything in the set, then it is true of any particular thing in the set.

Predicates

- A **predicate** is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables.

- The **domain** of a predicate variable is the set of all values that may be substituted in place of the variable.
 - If $D = \{x_1, x_2, \dots, x_n\}$, then:
 - $\forall x \in D, Q(x) \equiv Q(x_1) \wedge Q(x_2) \wedge \dots \wedge Q(x_n)$
 - $\exists x \in D, Q(x) \equiv Q(x_1) \vee Q(x_2) \vee \dots \vee Q(x_n)$

Operations

$\sim (\forall x \in D, P(x)) \equiv \exists x \in D, \sim P(x)$. Note negative of $\forall$ is $\exists$ and vice-versa.

- **Negation:**
 - $\sim (\forall x \in D, \exists y \in E \text{ such that } P(x, y)) \equiv \exists x \in D \text{ such that } \forall y \in E, \sim P(x, y)$
 - $\sim (\exists x \in D \text{ such that } \forall y \in E, P(x, y)) \equiv \forall x \in D, \exists y \in E \text{ such that } \sim P(x, y)$
- **Contrapositive/converse/inverse** of $\forall x \in D, (P(x) \rightarrow Q(x))$:
 - Contrapositive: $\forall x \in D, (\sim Q(x) \rightarrow \sim P(x))$
 - Converse: $\forall x \in D, (Q(x) \rightarrow P(x))$
 - Inverse: $\forall x \in D, (\sim P(x) \rightarrow \sim Q(x))$

Sufficiency/Necessity

- $\forall x, r(x)$ is a **sufficient condition** for $s(x)$ means $\forall x, (r(x) \rightarrow s(x))$.
- $\forall x, r(x)$ is a **necessary condition** for $s(x)$ means $\forall x, (\sim r(x) \rightarrow \sim s(x))$, or equivalently, $\forall x, (s(x) \rightarrow r(x))$.
- $\forall x, r(x)$ **only if** $s(x)$ means $\forall x, (\sim s(x) \rightarrow \sim r(x))$, or equivalently, $\forall x, (r(x) \rightarrow s(x))$.

Proofs

Assume we want to prove $p \rightarrow q$.

1. Direct proof / Counter-example
 - Suppose p .
 - *etc...*
 - Thus we get p . So $p \rightarrow q$.
2. Contraposition
 - Suppose $\sim q$.
 - *...etc*
 - Thus we get $\sim p$. So $\sim q \rightarrow \sim p \equiv p \rightarrow q$.
3. Contradiction
 - Suppose $\sim(p \rightarrow q)$
 - *...etc*
 - Contradiction
 - Hence $p \rightarrow q$.

- If we want to prove $p \leftrightarrow q$ then we have to prove both $p \rightarrow q \wedge q \rightarrow p$

Sets

- Set builder notation: $\{x \in U \mid P(x)\}$ meaning x must be in U and $P(x)$ is true
- Replacement notation: $\{t(x) \mid x \in A\}$ meaning the set of all $t(x)$ where $x \in A$
- Roster notation: $\{a, b, c, d, \dots\}$ just list it out

Symbol	Description
$\subseteq$	Subset: $A \subseteq B \Leftrightarrow \forall x(x \in A \Rightarrow x \in B)$
$\subsetneq$	Proper subset: $A \subsetneq B \Leftrightarrow A \subseteq B \wedge A \neq B$

$\not\subseteq$	Not subset: $A \not\subseteq B \Leftrightarrow \exists x(x \in A \wedge x \notin B)$
$\emptyset$	Empty set: the set with no elements, $\{\}$
$\cup$	Union: $A \cup B = \{x \in U : x \in A \vee x \in B\}$
$\cap$	Intersection: $A \cap B = \{x \in U : x \in A \wedge x \in B\}$
$\setminus$	Set difference: $B \setminus A = \{x \in U : x \in B \wedge x \notin A\}$
$\overline{A}$	Complement: $\overline{A} = \{x \in U : x \notin A\}$

- Ordered pairs: $(a, b) \equiv (x, y)$ iff $(a = x) \wedge (b = y)$
- Cartesian product: $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$

Intervals ($a \leq b$)

$$(a, b) = \{x : a < x < b\}, [a, b] = \{x : a \leq x \leq b\}$$

Indexed unions/intersections

$\bigcup_{i=0}^n A_i = \{x \in U \mid x \in A_i \text{ for some } i\}$, $\bigcap_{i=0}^n A_i = \{x \in U \mid x \in A_i \text{ for all } i\}$ (also for $i = 0$ to ∞).

Disjoint / partition

- Disjoint: $A \cap B = \emptyset$
- Mutually/pairwise disjoint: $A_i \cap A_j = \emptyset$ whenever $i \neq j$
- Partition of A : mutually disjoint sets whose union is A
- [Formally] $\mathcal{C}$ is a partition of A iff: all elements of $\mathcal{C}$ are non-empty subsets of A , and every $x \in A$ is in exactly one $S \in \mathcal{C}$, i.e. $\forall x \in A, \exists! S \in \mathcal{C} (x \in S)$

Power set

- $\mathcal{P}(A)$ = set of all subsets of A (power set axiom guarantees it's a set)
- $|A| = n \Rightarrow |\mathcal{P}(A)| = 2^n$

n-tuples

- $(x_1, \dots, x_n) = (y_1, \dots, y_n) \Leftrightarrow x_1 = y_1 \wedge \dots \wedge x_n = y_n$
- $A_1 \times \dots \times A_n = \{(a_1, \dots, a_n) : a_1 \in A_1 \wedge \dots \wedge a_n \in A_n\}$
- $A^n = A \times A \times \dots \times A$ (n times)

Subset relations

- $A \cap B \subseteq A, B$
- $A, B \subseteq A \cup B$
- $A \subseteq B \wedge B \subseteq C \rightarrow A \subseteq C$ (transitive)

Procedural definitions

- $a \in X \cup Y \Leftrightarrow a \in X \vee a \in Y$
- $a \in X \cap Y \Leftrightarrow a \in X \wedge a \in Y$
- $a \in X - Y \Leftrightarrow a \in X \wedge a \notin Y$
- $a \in \overline{X} \Leftrightarrow a \notin X$
- $(a, b) \in X \times Y \Leftrightarrow a \in X \wedge b \in Y$

Set identities

Same structure as logical equivalences ($\cup \sim \vee$, $\cap \sim \wedge$, $\emptyset \sim \text{false}$, $U \sim \text{true}$), and:

- Identity: $A \cup \emptyset = A$, $A \cap U = A$
- Complement: $A \cup \overline{A} = U$, $A \cap \overline{A} = \emptyset$
- Double complement: $\overline{\overline{A}} = A$

- Complements of $U, \emptyset$: $\overline{U} = \emptyset, \overline{\emptyset} = U$
- Set difference law: $A \setminus B = A \cap \overline{B}$

Relations

Definitions

- Relation from A to B : a subset $R \subseteq A \times B$
- xRy (x is R -related to y) iff $(x, y) \in R$

Properties

- Domain: $\text{Dom}(R) = \{a \in A : aRb \text{ for some } b \in B\}$
- Co-domain: $\text{coDom}(R) = B$
- Range: $\text{Range}(R) = \{b \in B : aRb \text{ for some } a \in A\}$

Operations

- Inverse relation R^{-1} from B to A : $R^{-1} = \{(y, x) \in B \times A : (x, y) \in R\}$; just invert the ordered pair
- Relation on a set A : relation from A to A , i.e. subset of $A \times A$
- Composition of relations: $R \subseteq A \times B, S \subseteq B \times C$, then $S \circ R \subseteq A \times C$: $\forall x \in A, \forall z \in C, (x(S \circ R)z \Leftrightarrow \exists y \in B (xRy \wedge ySz))$
 - Apply R first then apply S .

Reflexivity, Symmetry, Transitivity

- Reflexive: $\forall x \in A, xRx$
- Symmetric: $\forall x, y \in A, xRy \Rightarrow yRx$
- Transitive: $\forall x, y, z \in A, (xRy \wedge yRz) \Rightarrow xRz$
- Transitive closure R^t of R : smallest transitive relation containing R
 - R^t is transitive; $R \subseteq R^t$; if S transitive and $R \subseteq S$, then $R^t \subseteq S$
 - i.e. add the smallest number of ordered pairs to the relation R s.t. transitivity is satisfied for all elements

Equivalence partitions

- Relation induced by partition $\mathcal{C}$ of A : $\forall x, y \in A, xRy \Leftrightarrow \exists \text{ component } S \in \mathcal{C} \text{ s.t. } x, y \in S$
 - i.e. for all components of partition, relation forms when choosing all permutations of 2 plus pairing with each element with itself of each component
 - This relation satisfies reflexivity, symmetry and transitivity
 - Hence, $S = S^{-1}; S \circ S^{-1} = S$
- Equivalence relation: relation R on A that is reflexive, symmetric, transitive (R is usually written with $\sim$)
- Equivalence class of a under $\sim$: $[a]_{\sim} = \{x \in A : a \sim x\}$
 - i.e. basically a set where all elements have an equivalence relation to each other.
 - Note that the class has a as well as all other elements related to it
- Congruence mod n : $a \equiv b \pmod{n} \Leftrightarrow a - b = nk$ for some $k \in \mathbb{Z}$, i.e. $n \mid (a - b)$
 - i.e. $a \bmod n = b \bmod n$, both leave the same remainder when divided by n .
- Quotient set: $A / \sim = \{[x]_{\sim} : x \in A\}$ (set of all equivalence classes)

Partial Order Relations

- Antisymmetric: relation R on A s.t. $\forall x, y \in A, (xRy \wedge yRx) \Rightarrow x = y$
 - ie. if x and y relate both ways, they must be the same element - no two **distinct** elements can relate to each other in both directions
- Partial order: relation R on A that is reflexive, antisymmetric, transitive (written $\preceq$)
- Poset: set A with partial order $\preceq$, denoted $(A, \preceq)$
- Hasse diagram: for distinct $x, y, m \in A$, if $x \preceq y$ and no m with $x \preceq m \preceq y$, draw x below y joined by a line; else no line
 - Minimal: no lines below it; Maximal: no lines above it
 - Minimum: a minimal element that is comparable to all elements; Maximum: a maximal element that is comparable to all elements. Note that they are unique if they exist.
- Comparable: x, y comparable iff $x \preceq y$ or $y \preceq x$; else noncomparable

Total Order Relations

- Total order: partial order R on A s.t. $\forall x, y \in A, xRy \vee yRx$ (every pair comparable)
- Linearization: total order $\preceq^*$ on A extending partial order $\preceq$, i.e. $\forall x, y \in A, x \preceq y \Rightarrow x \preceq^* y$
 - eg. on the hasse diagram,
 1. find any minimal/the minimum element and push it to a list. Remove the element.
 2. repeat step 1 until exhausted.
 3. stack contains linearization. $x \preceq^* y$ iff x was appears earlier than y .
- Well-ordered: total order $\preceq$ on A s.t. every non-empty subset has a smallest element: $\forall S \in \mathcal{P}(A), S \neq \emptyset \Rightarrow (\exists x \in S, \forall y \in S, x \preceq y)$

Functions

- Function $f : X \rightarrow Y$: relation s.t. (F1) $\forall x \in X, \exists y \in Y, (x, y) \in f$ and (F2) that y is unique
 - ie. every x maps to exactly one y
- $f(x) = y$: means $(x, y) \in f$; x is the **argument**, y is the **output/image** of x ; x is a **preimage** of y

Setwise image and preimage

- Setwise image: for $A \subseteq X, f(A) = \{f(x) : x \in A\}$
- Setwise preimage: for $B \subseteq Y, f^{-1}(B) = \{x \in X : f(x) \in B\}$
 - ie. $f(A)$ = map f over A ; $f^{-1}(B)$ = elements mapping into B

Sequences and Strings

- Sequence: $a_0, a_1, a_2, \dots$ represented by function $a : \mathbb{Z}_{\geq 0} \rightarrow \text{codomain}$ with $a(n) = a_n$
- Fibonacci: $F_0 = 0, F_1 = 1, F_{n+2} = F_{n+1} + F_n$
- String over A : $a_0 a_1 \dots a_{l-1}, a_i \in A$, length $l \in \mathbb{Z}_{\geq 0}$; empty string ε has $l = 0$
- Sequence equality: $a = b$ iff $a(n) = b(n) \forall n \in \mathbb{Z}_{\geq 0}$; set of all sequences over A is A^∞ or $\text{Seq}(A)$
- String equality: $s_1 = s_2$ iff same length and $a_i = b_i \forall i$; set of all strings over A is A^* or $\text{Str}(A)$
- Function equality: $f = g$ iff same domain, same codomain, and $f(x) = g(x) \forall x$

Injection and Surjection

- Injective (one-to-one): $\forall x_1, x_2 \in X, f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ (equiv. $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$)
- Surjective (onto): $\forall y \in Y, \exists x \in X, y = f(x)$
 - ie. every codomain element has a preimage, so range = codomain